THE AMBAR EVIDENCE GUIDE

The Science of Plasma Exchange for Alzheimer's Disease

A synthesis of the landmark AMBAR trial, its independent validation, and what the evidence means for patients and families. Compiled from the four-part series at allenpgreenmd.com.

52–71%
Less cognitive & functional decline vs. placebo

322
Patients evaluated across 41 clinical sites

Phase 2b/3
Randomized, placebo-controlled trial

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THE TRIAL

What Is the AMBAR Trial?

The AMBAR study (Alzheimer's Management By Albumin Replacement) is the largest randomized controlled trial ever conducted on therapeutic plasma exchange for Alzheimer's disease. It was a Phase 2b/3, multicenter, double-blind, placebo-controlled trial across 41 clinical sites in Spain and the United States. Of 347 patients randomized, **322 were evaluated** (242 received plasma exchange, 80 sham placebo).

Patients received two phases of treatment: six weeks of weekly conventional TPE, followed by 12 months of monthly low-volume plasma exchange, with albumin as the replacement fluid. A sham procedure served as the placebo control, an important design feature that minimized bias in a procedural trial. Notably, the study enrolled on clinical diagnosis alone and did not require amyloid-positive biomarker confirmation, a detail that would prove scientifically significant.

> **How TPE works against Alzheimer's:** Alzheimer's involves accumulation of amyloid-β, inflammatory cytokines, oxidized lipids, and other neurotoxic factors in blood and brain, which stay in equilibrium across the blood-brain barrier. TPE removes the full spectrum of these circulating factors in a single procedure and replaces them with 5% albumin, which itself binds and neutralizes amyloid-β and has anti-inflammatory and antioxidant properties.

52%	66%	71%	100%
Less functional decline	**Less cognitive decline**	**Less dementia progression**	**Less global decline**
ADCS-ADL · p = .03	ADAS-Cog · p = .06	CDR-SB · p = .002	ADCS-CGIC · p < .0001

Effect sizes vs. placebo, full trial population. Boada M, et al. Alzheimer's & Dementia. 2020;16(10):1412–1425.

Disease Severity: The Subgroup Findings

Pre-specified analyses by baseline MMSE revealed clinically important differences by disease stage.

MODERATE ALZHEIMER'S · MMSE 18–21

61% less decline

- ADCS-ADL: 61% less decline (p = .002)
- ADAS-Cog: 61% less decline (p = .05)
- CDR-SB: 50–60% less decline (p = .01–.04)
- ADCS-CGIC: 57–71% less decline (p = .002–.02)

MILD ALZHEIMER'S · MMSE 22–26

Improved from baseline

- On global measures, treated patients improved from baseline while placebo declined, with **effect sizes of 157–171% for CDR-SB** (p = .02–.04) and **200% for ADCS-CGIC** (p = .004–.04).
- These effect sizes exceed 100% because treated patients did not merely decline more slowly; they saw actual improvement on some measures.
- Standard cognitive tests (ADAS-Cog) showed no change, likely a ceiling effect.

INDEPENDENT VALIDATION

Real-World Replication: The Argentina Study

A common concern about AMBAR was that it was a single trial sponsored by Grifols, the manufacturer of therapeutic albumin. In 2025, that concern was answered. An independent research team in Buenos Aires published a real-world cohort study: **32 patients** who received TPE compared to **194 historical controls** from the same memory clinic, using a modified AMBAR protocol conducted entirely via peripheral venous access, no central lines.

Measure	Result	P-value
MMSE (overall cognition)	45% less decline	$p < .001$
Immediate memory recall (RAVLT)	88% less decline	$p < .001$
Delayed memory recall (RAVLT)	74% less decline	$p = .04$
Recognition memory	69% less decline	$p < .001$
Executive function (phonemic fluency)	49% less decline	$p < .001$
Semantic fluency	37% less decline	$p < .001$

Taragano F, et al. A real-world study on the safety and efficacy of therapeutic plasma exchange in patients with Alzheimer's disease. Journal of Alzheimer's Disease. 2025;108(1):129–141.

Why independent replication matters: Different investigators, country, patient population, and cognitive tests. A modified protocol, done via peripheral access rather than central lines. Yet the magnitude of benefit was remarkably consistent with AMBAR, a 45% reduction in MMSE decline closely mirroring AMBAR's 52% effect on ADCS-ADL. When two independent studies using different methods show similar results, that is compelling evidence.

HOW TPE COMPARES

TPE vs. FDA-Approved Alzheimer's Drugs

The following comparison was presented at the ASFA 2024 conference by the Ace Alzheimer Center Barcelona. It places AMBAR's results alongside FDA-approved anti-amyloid therapies across three primary outcome measures and ARIA (brain swelling/bleeding) risk.

Treatment	ADCS-ADL	ADAS-Cog	CDR-SB	ARIA Risk
Aducanumab (ENGAGE)	18%	11%	-2%	40%
Aducanumab (EMERGE)	40%	27%	22%	40%
Lecanemab (Leqembi)	37%	26%	27%	12.6–14%
Donanemab (Kisunla)	40%	32%	37%	24–31%
AMBAR (TPE)	**52%**	**66%**	**71%**	**0%**

Source: Ace Alzheimer Center Barcelona, ASFA 2024 Conference. "Alzheimer's disease, current treatments and the role of plasma exchange." February 7, 2024.

> Across all three outcome measures, AMBAR's effect sizes met or exceeded every FDA-approved anti-amyloid drug, and did so with **no ARIA risk**, because TPE is not an anti-amyloid antibody and does not carry the brain-swelling and micro-hemorrhage risk that requires ongoing MRI monitoring with the antibody therapies.

THE BREAKTHROUGH FINDING

TPE Works Regardless of Amyloid Status

Perhaps the most scientifically significant AMBAR finding did not appear in the original 2020 publication. Because AMBAR enrolled on clinical diagnosis alone, later analysis of CSF samples found that roughly **30% of enrolled patients were amyloid-negative**, without the hallmark pathological feature of Alzheimer's. The critical result: **both amyloid-positive and amyloid-negative patients showed similar clinical benefit**, with amyloid-negative patients improving significantly on ADAS-Cog, CDR-SB, and ADCS-CGIC versus controls.

BEYOND THE AMYLOID HYPOTHESIS

If TPE worked solely by removing amyloid, it should not help amyloid-negative patients. That it does implicates other mechanisms: inflammation, cellular senescence, and oxidative stress.

AN UNMET NEED ADDRESSED

FDA-approved drugs require amyloid-positive status. Patients with an Alzheimer's phenotype but negative amyloid have no approved options, yet benefited equally from TPE.

A MULTI-FACTORIAL MECHANISM

Alzheimer's appears multi-factorial. A treatment addressing the full inflammatory and aging milieu may outperform targeting a single pathway.

> *"The AMBAR trial showed that plasma exchange with albumin replacement can meaningfully slow the progression of Alzheimer's disease. Across each clinical measure tested, the effect size exceeded anything an FDA-approved drug has produced in the same patient population."*
>
> — Dr. Allen P. Green, MD · Associate Medical Director, Global Apheresis

The Treatment Protocol

Phase	Duration	What happens
Intensive	Weeks 1–6 (6 sessions)	Weekly full-plasma-volume exchange; plasma removed and replaced with 5% albumin. Rapidly reduces circulating amyloid-β, inflammatory cytokines, and other neurotoxic factors.
Maintenance	Months 2–14 (monthly)	Monthly lower-volume exchanges that sustain the benefit and prevent re-accumulation of harmful plasma factors. AMBAR maintained patients for 12 months in this phase.

THE ADOPTION GAP

If It Works, Why Hasn't Your Doctor Recommended It?

Published in *Alzheimer's & Dementia* in 2020, with effect sizes larger than any approved drug and no MRI monitoring required, and yet most neurologists have never discussed it. Understanding why is as important as understanding the science.

No patent, no marketing budget. Plasma exchange uses albumin, a long-generic product. No company holds exclusive rights, so there is no industry incentive to fund the education campaigns that drive physician awareness.

Regulatory pathway. TPE for Alzheimer's is off-label in the US. FDA approval requires a sponsor to file and fund the application, and no single company has sufficient commercial incentive.

Implementation complexity. A plasma exchange protocol needs specialized equipment, trained nursing staff, and clinical infrastructure most neurology practices do not have. It means referring outside the practice.

Specialty silo. Therapeutic apheresis sits within hematology and transfusion medicine, not neurology. Most neurologists have limited exposure to the literature.

NEXT STEP

Understanding If TPE Is Right for Your Situation

The evidence is the foundation. The next step is an individual conversation about candidacy, what the protocol looks like in practice, and whether therapeutic plasma exchange makes sense for you.

Read the full series

allenpgreenmd.com/ambar-series

The complete four-part analysis: trial design, results, independent validation, drug comparisons, and the amyloid-negative finding.

Read the clinical overview

allenpgreenmd.com/tpe-alzheimers

Candidacy criteria, full treatment protocol, and how TPE fits alongside other Alzheimer's treatments.

Book a discovery call

globalapheresis.com/free-consultation

A complimentary conversation with the clinical team about your goals and whether TPE may be appropriate.

PRIMARY REFERENCES

References

1. Boada M, López OL, Olazarán J, et al. A randomized, controlled clinical trial of plasma exchange with albumin replacement for Alzheimer's disease: Primary results of the AMBAR Study. *Alzheimer's & Dementia.* 2020;16(10):1412–1425. doi:10.1002/alz.12137
2. Taragano F, Seinhart D, Epstein P, et al. A real-world study on the safety and efficacy of therapeutic plasma exchange in patients with Alzheimer's disease. *Journal of Alzheimer's Disease.* 2025;108(1):129–141. doi:10.1177/13872877251375430
3. Costa M, Boada M. Letter to the Editor: AMBAR — A Therapeutical Approach for Alzheimer's Disease Patients Regardless of Amyloid Status. *The Journal of Prevention of Alzheimer's Disease.* 2023;10(1):148–149. doi:10.14283/jpad.2022.100 [Amyloid-negative subgroup finding]
4. Ace Alzheimer Center Barcelona. "Alzheimer's disease, current treatments and the role of plasma exchange." ASFA 2024 Conference Webinar, February 7, 2024. [Source for the comparative efficacy table]